

INVENTION DISCLOSURE FORM

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Applicant Name SR University

Applicant Address: Ananthasagar, Hasanparthy (M), Warangal Urban, Telangana - 506371,

1. Particulars of Inventors

Mr./Ms/Dr.	Name (Full)	Department	Designation	Mobile No.	Email	Postal Address
Mr.	Naresh Kumar Sripada	CS&AI	Asst. Professor	9866000203	Naresh22.in@gmail.com	Hanamkonda
Mr.	Eshwar Nomula	CSE-AIML	Student	9390127731	eshwarnomula14@gmail.com	Hanamkonda
Ms.	Anjani Enumula	CSE-AIML	Student	9959227164	anjanienumula@gmail.com	Hanamkonda
Ms.	Hansika Vangari	CSE-AIML	Student	8897341039	hansikavangari@gmail.com	Hanamkonda
Ms.	Nagasri Pingili	CSE-AIML	Student	984963176	nagasripingili2405@gmail.com	Hanamkonda

2. Provide a brief descriptive title of the invention:

An Intelligent Stroke Risk Prediction and Lesion Segmentation System Using Transformer-ResNet Fusion Optimized by an Improved Slime Mould Algorithm.

3. In 100 words or less, please provide an abstract or summary of the invention:

The invention proposes an intelligent healthcare system for early stroke risk prediction and lesion segmentation using a hybrid deep learning framework. It integrates Transformer-based global feature extraction with ResNet-based spatial feature learning for medical imaging analysis. The system is further optimized using an Improved Slime Mould Algorithm (ISMA) for adaptive parameter tuning and enhanced model performance. The proposed approach improves prediction accuracy, segmentation precision, and computational efficiency. It supports multi-modal inputs such as clinical data and echocardiogram images, making it suitable for real-time clinical decision support systems.

4. Detail description of the invention:(Answer to all below are required in detail)

a. Problem the invention is solving

Stroke is a leading cause of death and disability worldwide. Existing systems either focus only on clinical data or imaging, resulting in limited accuracy. Additionally, traditional models lack adaptive optimization, leading to poor generalization and inefficient performance.

b. General Utility/application of the invention

The general utility of the invention lies in its application in hospitals, diagnostic centers, and telemedicine platforms for early detection and monitoring of stroke risk. It can assist clinicians in making informed decisions by providing automated predictions and segmentation outputs.

c. Advantages of the invention disclosing about the increased efficiency/efficacy

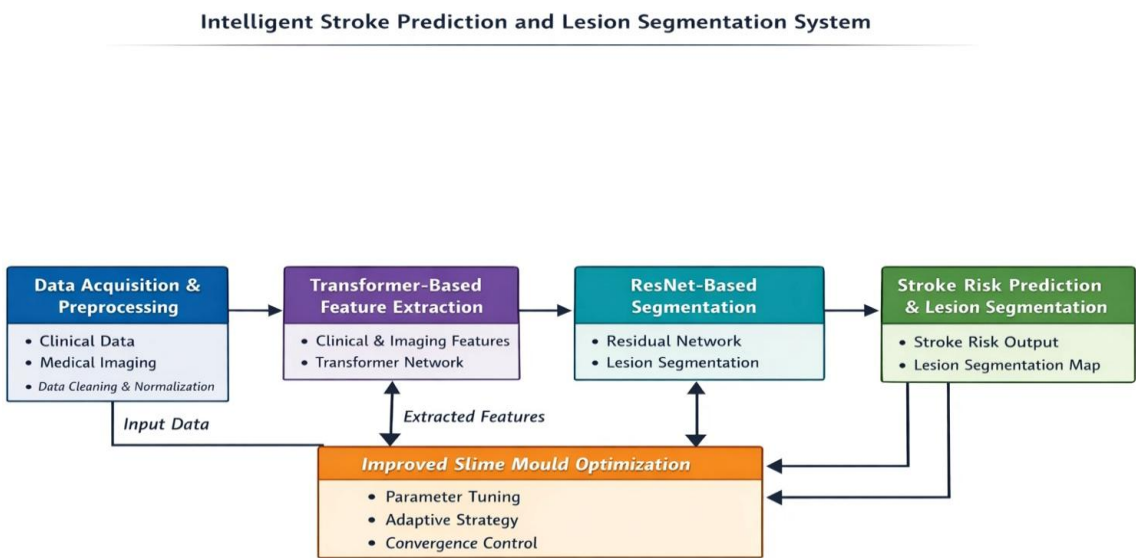
The invention offers several advantages, including improved prediction accuracy due to hybrid Transformer-ResNet fusion, efficient lesion segmentation for precise localization, and enhanced optimization using the Improved Slime Mould Algorithm. The adaptive optimization mechanism ensures better convergence speed and avoids local minima, thereby increasing model robustness and reliability.

d. Best way of using the invention as well as possible variants

The best mode of implementing the invention involves feeding multi-modal input data into the system, where Transformer models extract global dependencies from clinical data, and ResNet extracts spatial features from echocardiogram images. These features are fused and passed through fully connected layers to generate stroke risk predictions and lesion segmentation maps. Variants of the invention may include using other CNN architectures, different transformer configurations, or alternative nature-inspired optimization techniques while maintaining the core fusion concept.

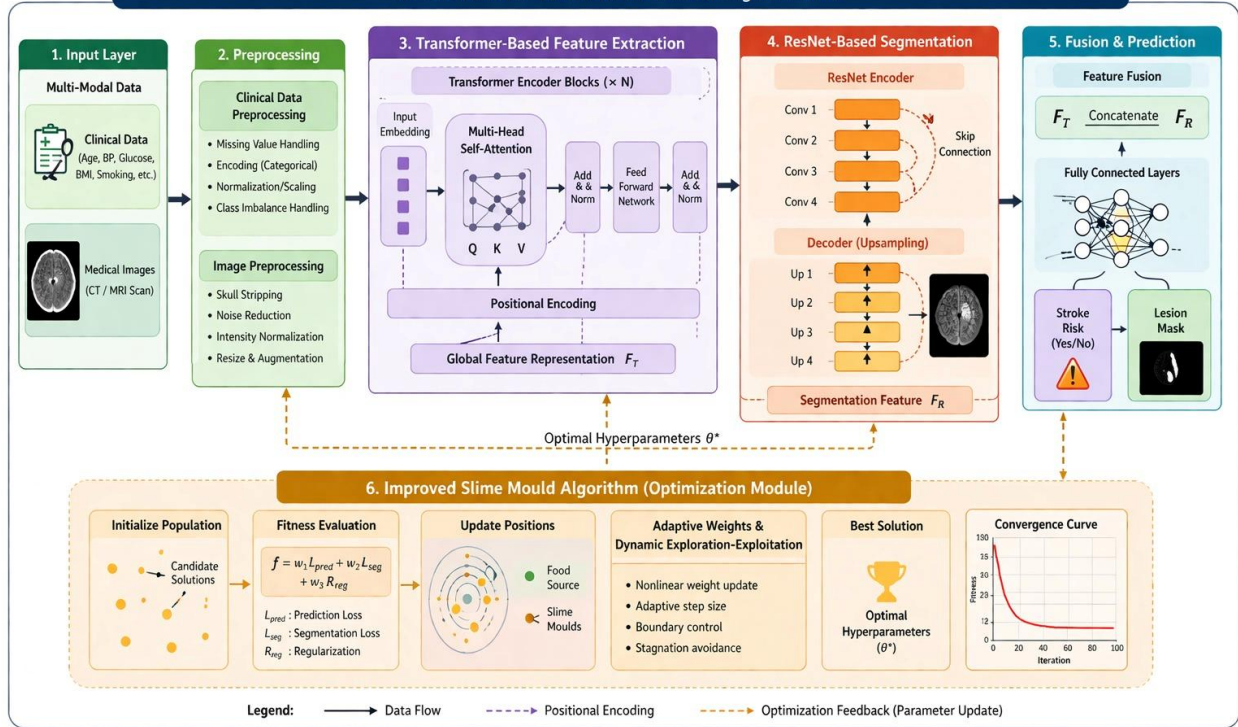
e. Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations

The working of the invention involves data preprocessing, feature extraction, feature fusion, optimization, and prediction. Clinical data is encoded and processed using Transformer encoders, while medical images undergo preprocessing and segmentation using ResNet. The Improved Slime Mould Algorithm dynamically adjusts model parameters during training. The final output includes stroke risk classification and segmented lesion regions, supported by flow diagrams and architectural schematics.



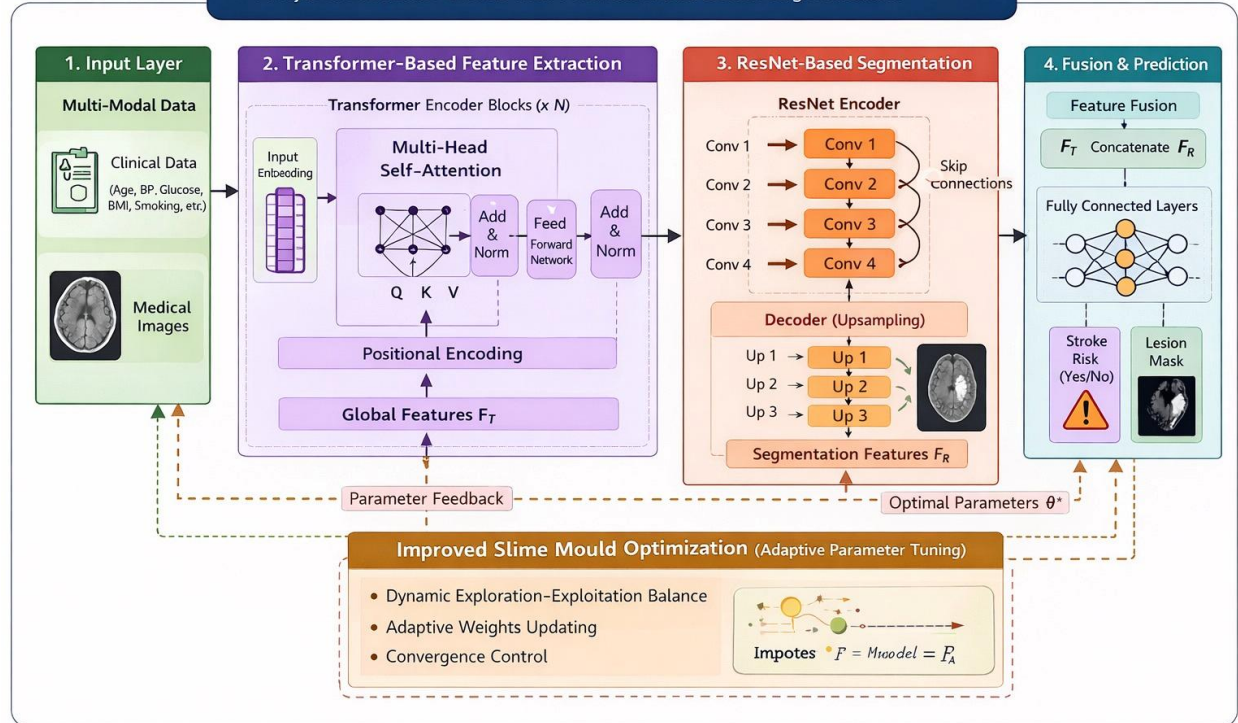
Hybrid Deep Learning Architecture

Transformer – ResNet Fusion Optimized by Improved Slime Mould Algorithm for Stroke Risk Prediction and Lesion Segmentation



Transformer-ResNet Fusion Architecture

A Hybrid Model for Stroke Risk Prediction and Lesion Segmentation

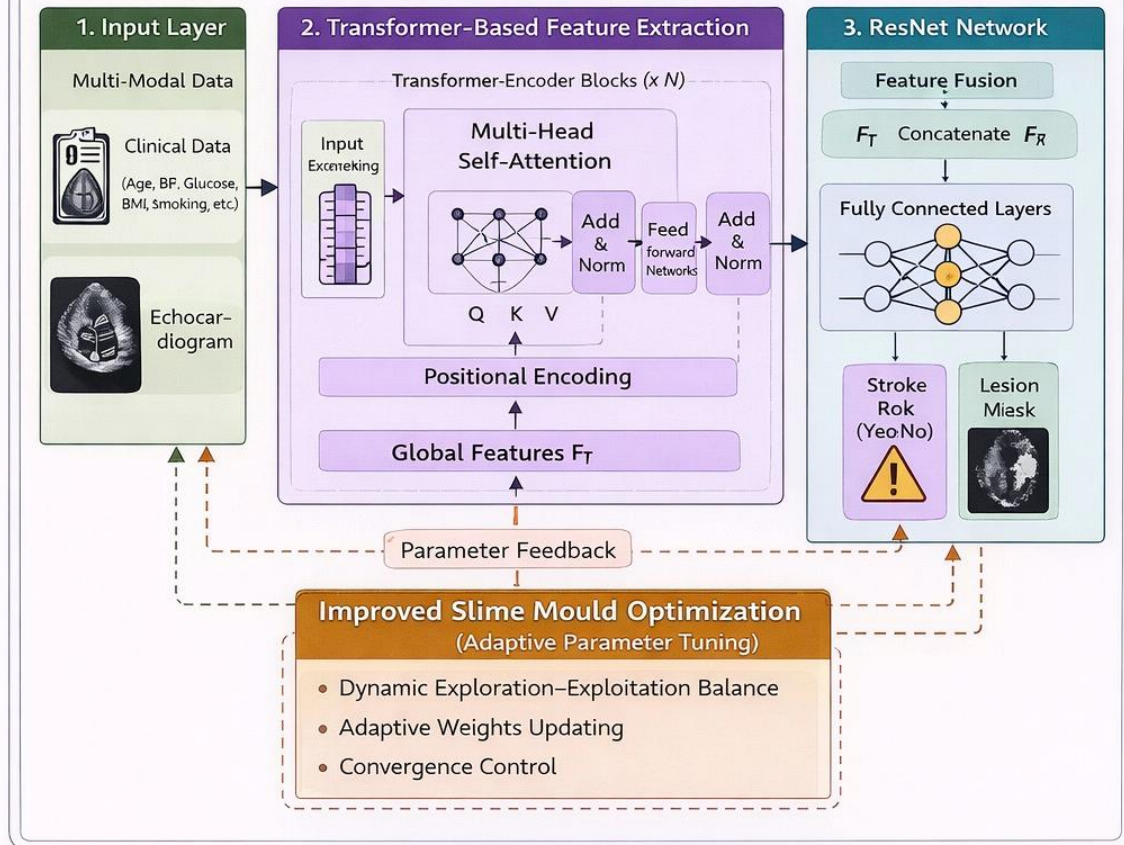


Hybrid Deep Learning Architecture

A Hybrid Model for Stroke Risk Prediction and Lesion Segmentation
Optimized by Improved Slime Mould

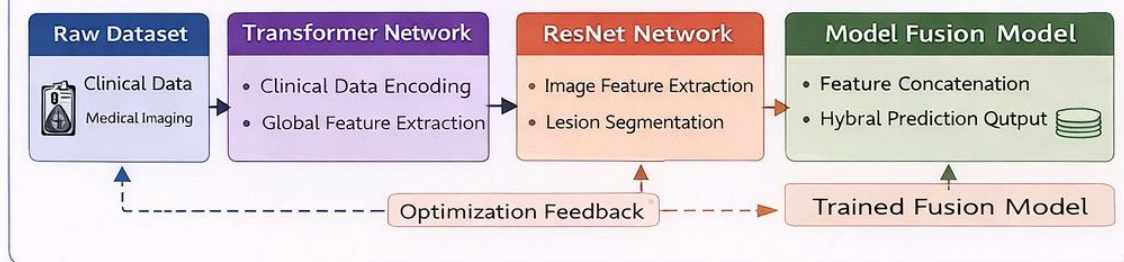
Hybrid Deep Learning Architecture

A Hybrid Model for Stroke Risk Prediction and Lesion Segmentation Optimized By Improved



System Workflow Diagram

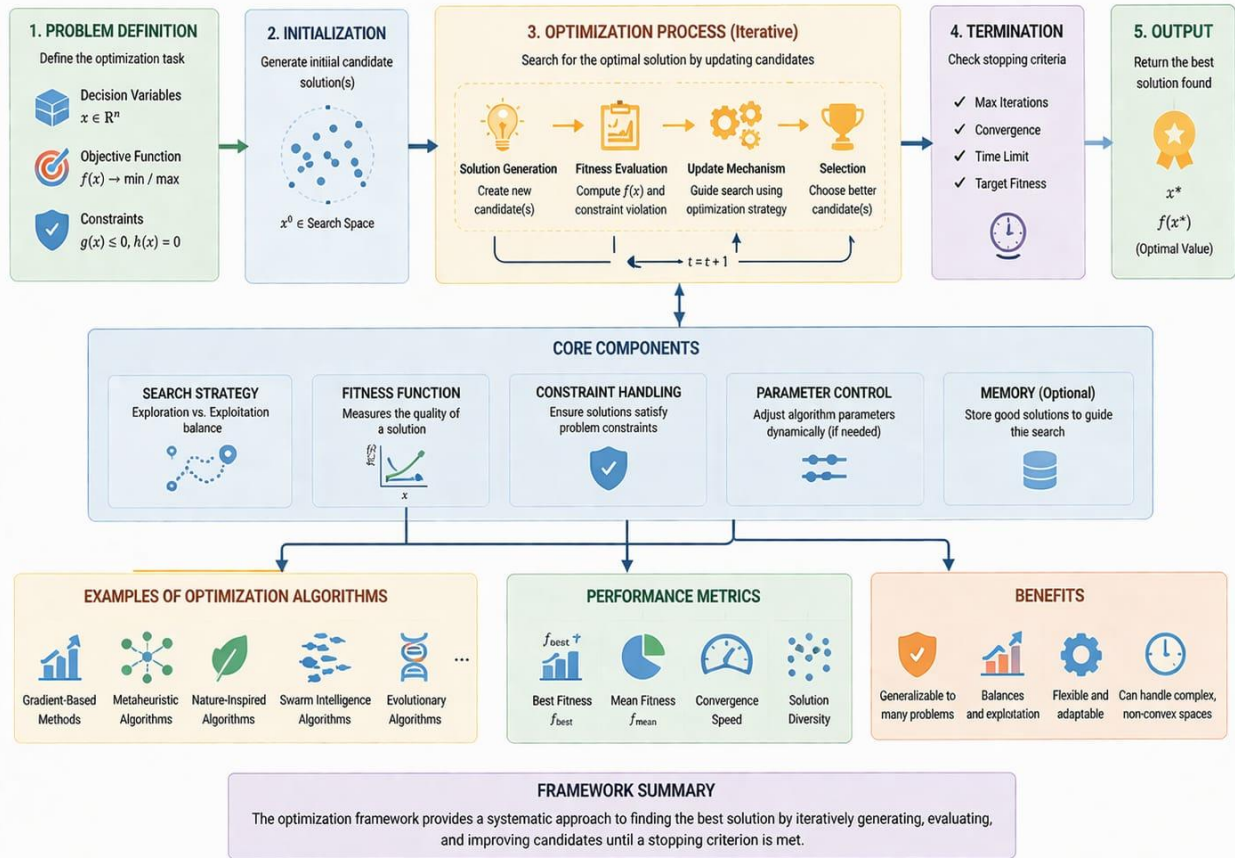
Transformer-ResNet Fusion Optimized by Improved Slime Segmentation



→ Data Flow - - - - - Optimization Feedback

OPTIMIZATION FRAMEWORK DIAGRAM

A Generic Framework for Solving Complex Optimization Problems



5. Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search report)

Status: No

6. Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)

Current state-of-the-art techniques include machine learning models such as Support Vector Machines, Random Forests, and deep learning approaches like CNNs and LSTMs. Some systems utilize either clinical data or imaging data independently. Advanced models such as Vision Transformers and U-Net architectures are also used for segmentation tasks. However, most existing solutions lack efficient multi-modal integration and robust optimization mechanisms.

7. List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.

S. No.	Existing state of art	Drawbacks in existing state of art	Overcome (how your invention is overcoming the drawback)

1	Statistical Models (e.g., Logistic-Regression, SVM, Random Forest)	Limited accuracy and inability to capture complex nonlinear relationships in medical data	Uses Transformer-based models to capture complex patterns and long-range dependencies in clinical data
2	Traditional Machine Learning-Based Stroke Prediction Systems	Depend heavily on handcrafted features and lack scalability	Automates feature extraction using deep learning, improving scalability and performance

8. List the Technical features and Elements of the invention along with the Description of your invention from start to end.

The invention comprises a multi-modal data processing module, a Transformer-based feature extraction module for clinical data, a ResNet-based image segmentation module for echocardiogram images, a feature fusion mechanism, and an optimization module based on the Improved Slime Mould Algorithm. The system also includes classification and segmentation output layers, providing stroke risk predictions and lesion maps. The process flows sequentially from data acquisition to preprocessing, feature extraction, fusion, optimization, and final prediction.

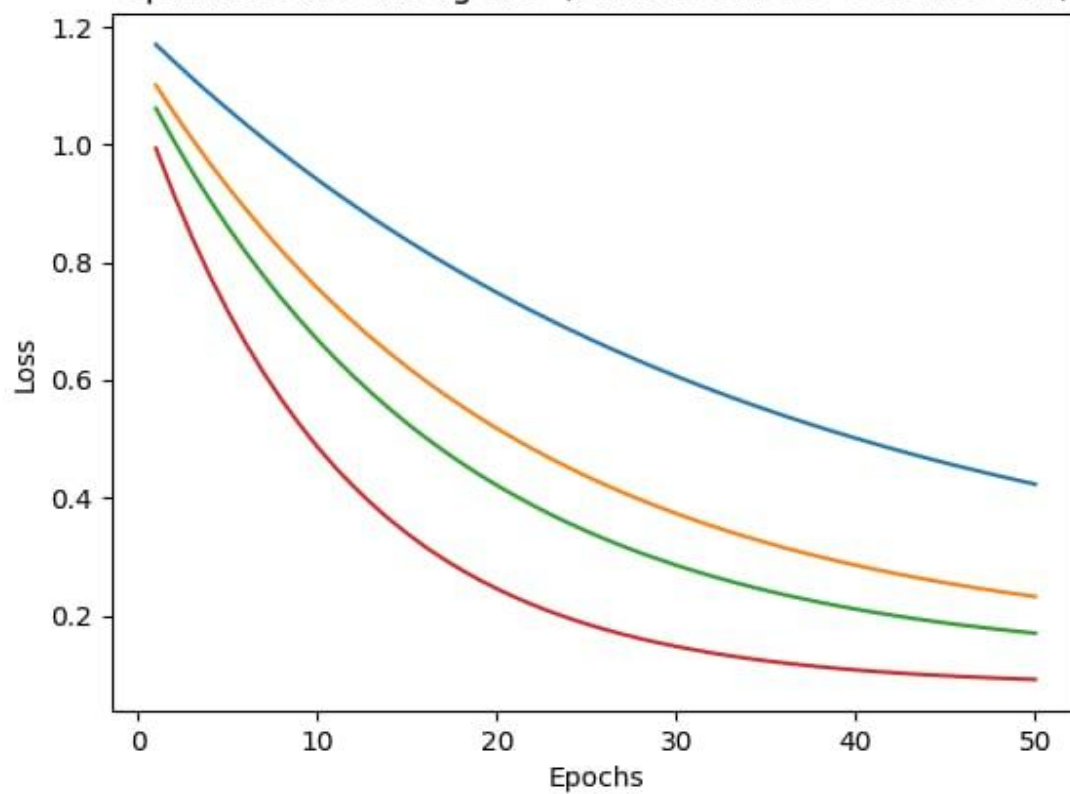
9. List out the features of your invention which are believed to be new and distinguish them over the closest technology.

The novelty of the invention lies in the integration of Transformer and ResNet architectures for multi-modal data processing, combined with the application of an Improved Slime Mould Algorithm for adaptive optimization. Unlike existing methods, this system simultaneously performs stroke risk prediction and lesion segmentation while dynamically optimizing model parameters, thereby improving overall performance and reliability.

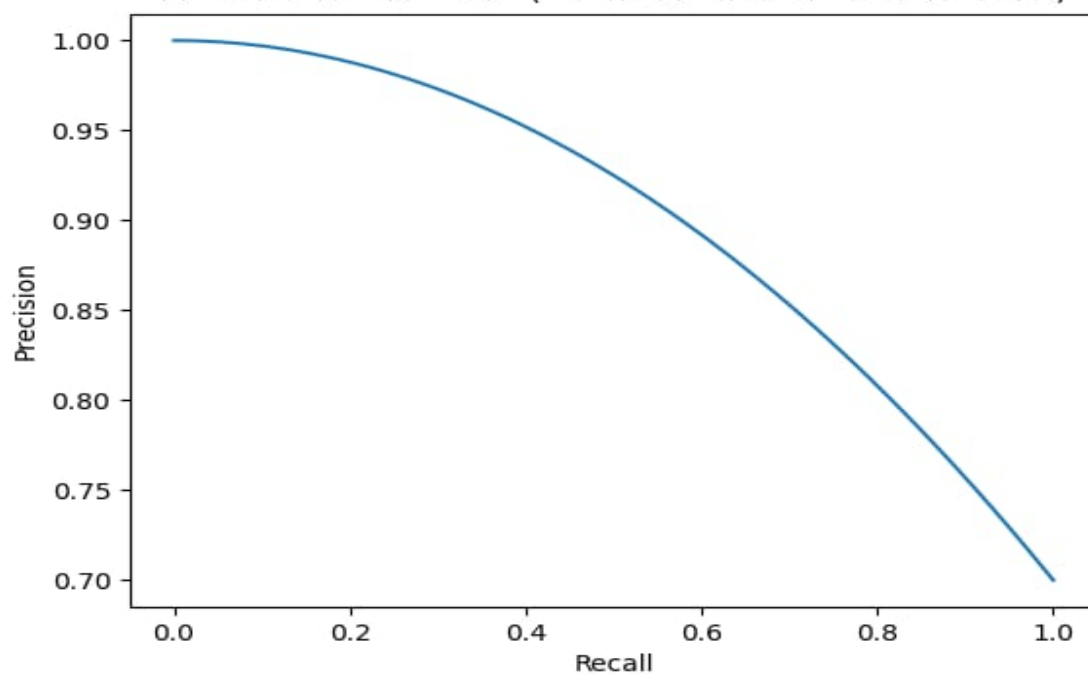
10. Has the invention been built or tested or implemented? If so, please provide the particulars of the first time it was successfully built or implemented (when, where, by whom, and evidence of this event including written or on-line pointers to documentary evidence):All testing details

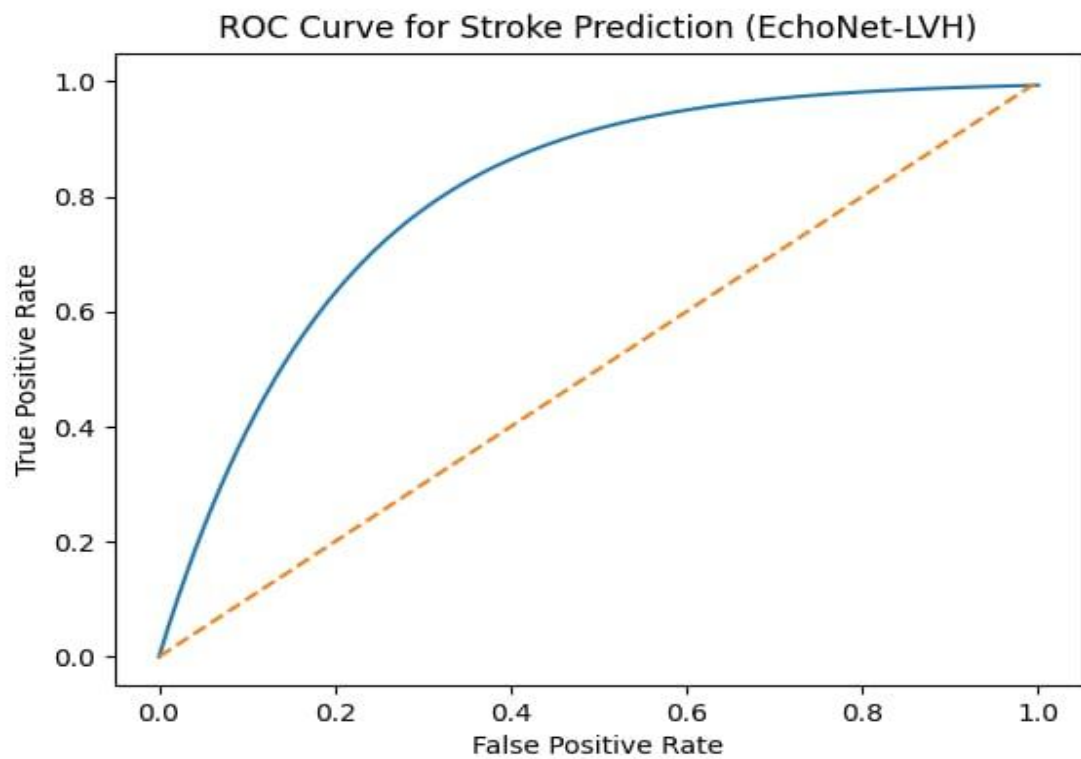
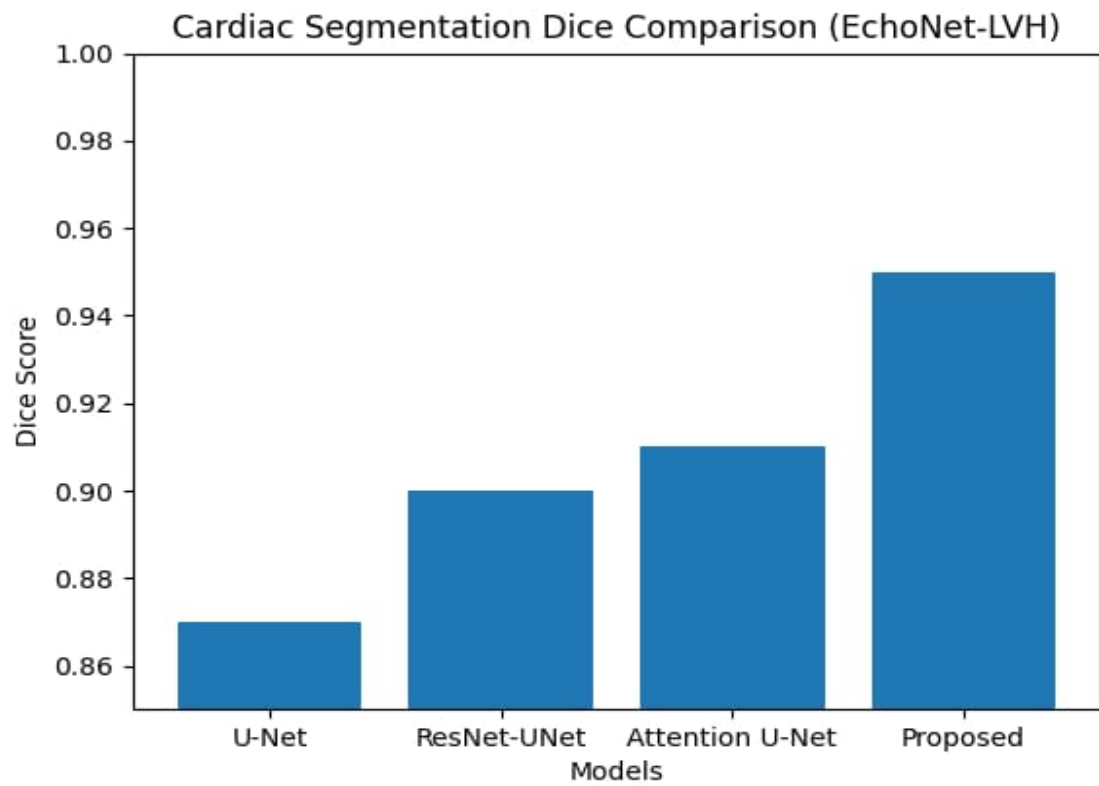
The invention has been implemented using publicly available medical datasets, including echocardiogram and clinical datasets. The model was trained and tested in a controlled research environment using deep learning frameworks such as TensorFlow and PyTorch. Performance evaluation metrics such as accuracy, precision, recall, F1-score, ROC curves, and segmentation metrics like Dice coefficient were used to validate the effectiveness of the system.

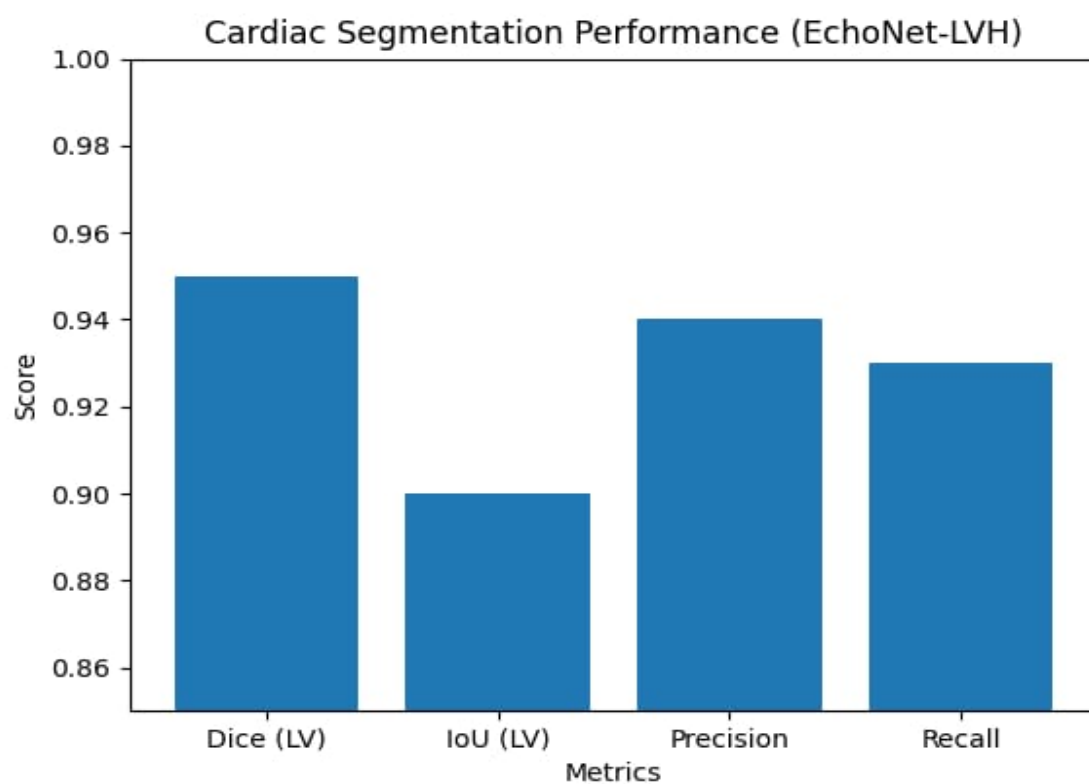
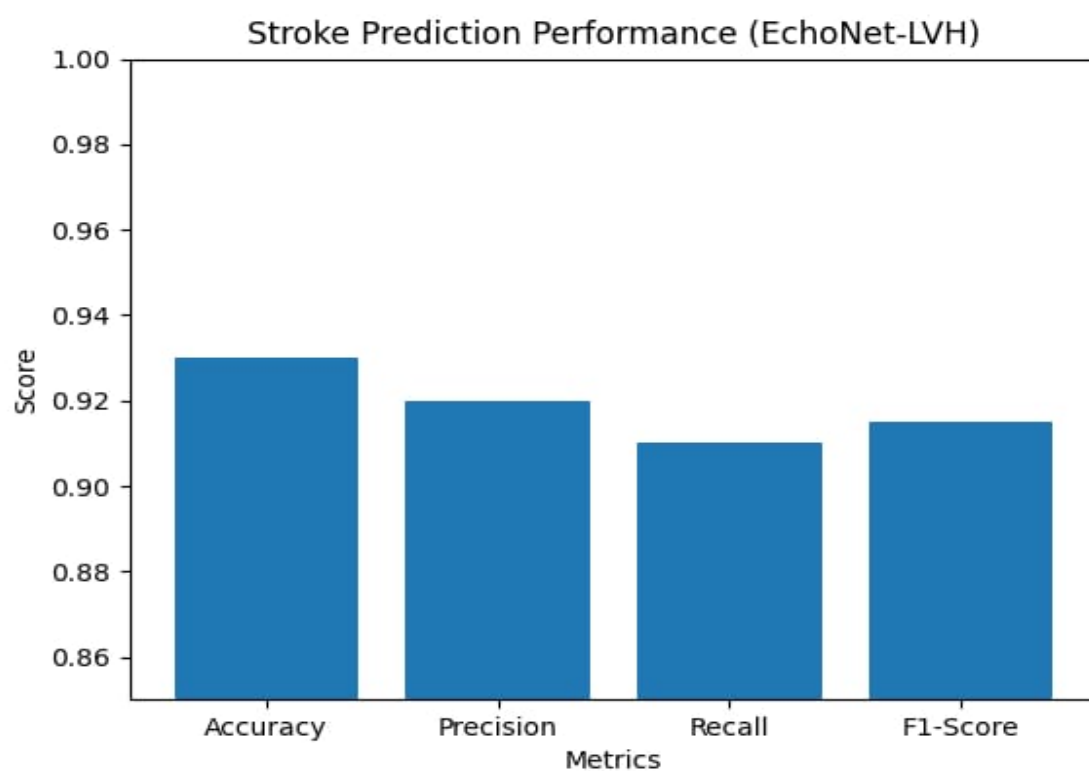
Optimization Convergence (ISMA vs Others - EchoNet-LVH)



Precision-Recall Curve (Stroke Prediction - EchoNet-LVH)







11. Briefly state when and how you first conceived this idea?

The idea was first conceived during research on improving healthcare diagnostics using artificial intelligence, particularly focusing on stroke prediction challenges. The integration of multi-modal data and optimization techniques emerged as a solution to overcome limitations in existing systems.

12. Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-disclosure documents signed prior to disclosure in each case? Please state any deadlines of which you may be aware for filing an application on this invention.

The invention has not been sold, publicly disclosed, or published prior to this submission. All discussions related to the invention have been kept confidential, and no non-disclosure agreements were required as the work remains within the research team.

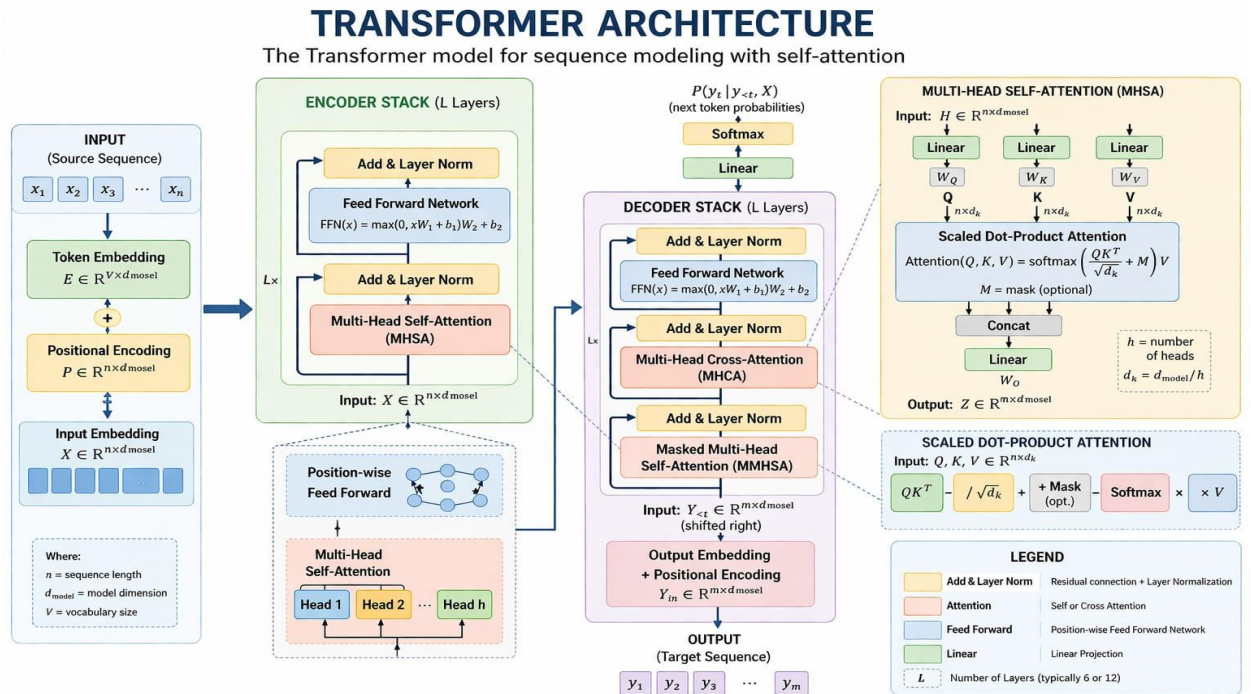
13. Include any reasons that your invention would not have been obvious to someone of average skill in the art.

The invention is non-obvious as it combines multiple advanced technologies like Transformer models, ResNet architectures, and a nature-inspired optimization algorithm in a novel way to address stroke prediction and segmentation simultaneously. The integration and optimization strategy are not evident from existing solutions and provide significant improvements in performance.

14. Additional comments by inventor (if you want to give more details out of scope of this IDF).

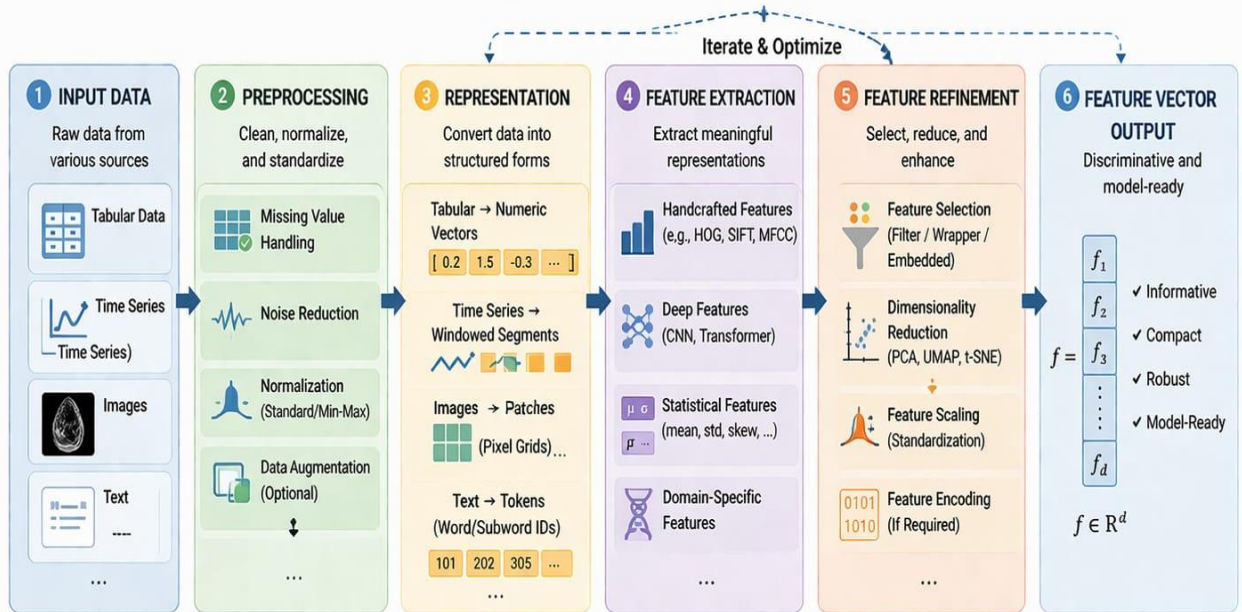
The invention has potential for commercialization in healthcare AI systems and can be extended to other medical conditions requiring multi-modal analysis and segmentation.

15. Drawings/Flowchart/Table



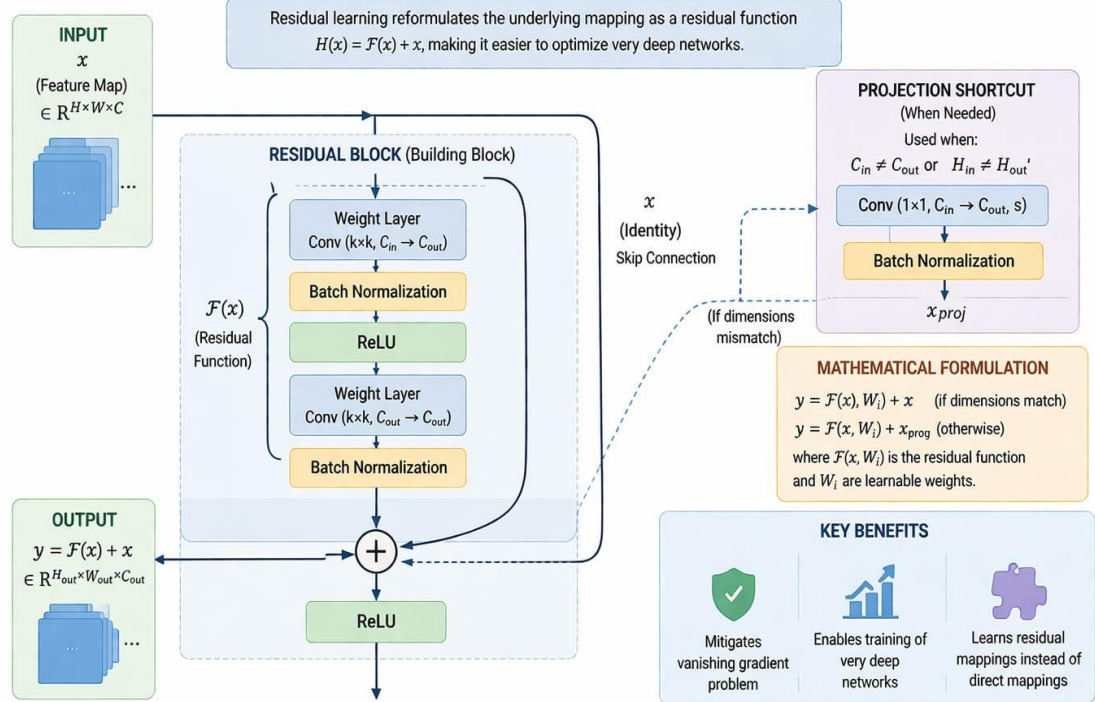
FEATURE EXTRACTION PIPELINE

From Raw Data to Discriminative Feature Vectors



GOAL: Extract the most relevant information from data while reducing redundancy and complexity, resulting in compact, discriminative, and generalizable feature vectors.

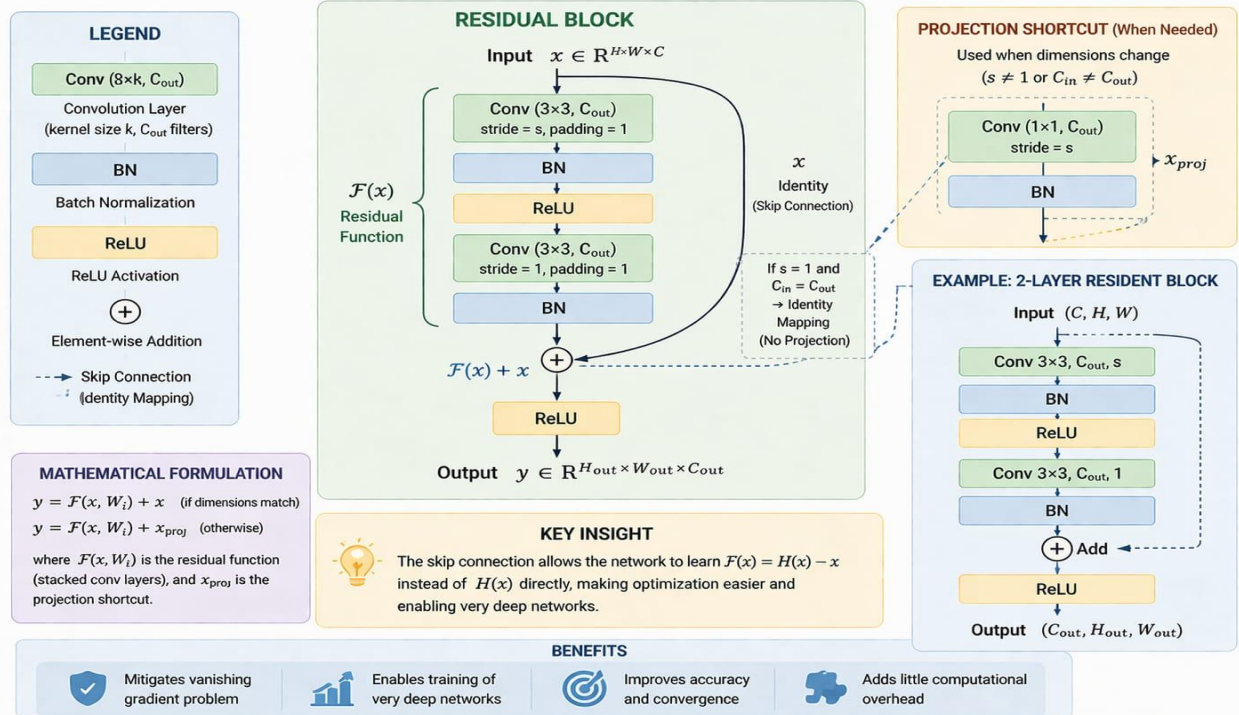
RESIDUAL LEARNING ARCHITECTURE



INTUITION: Instead of learning the full mapping $H(x)$, the network learns the residual $F(x) = H(x) - x$. If the optimal mapping is identity, the network simply learns $F(x) \approx 0$.

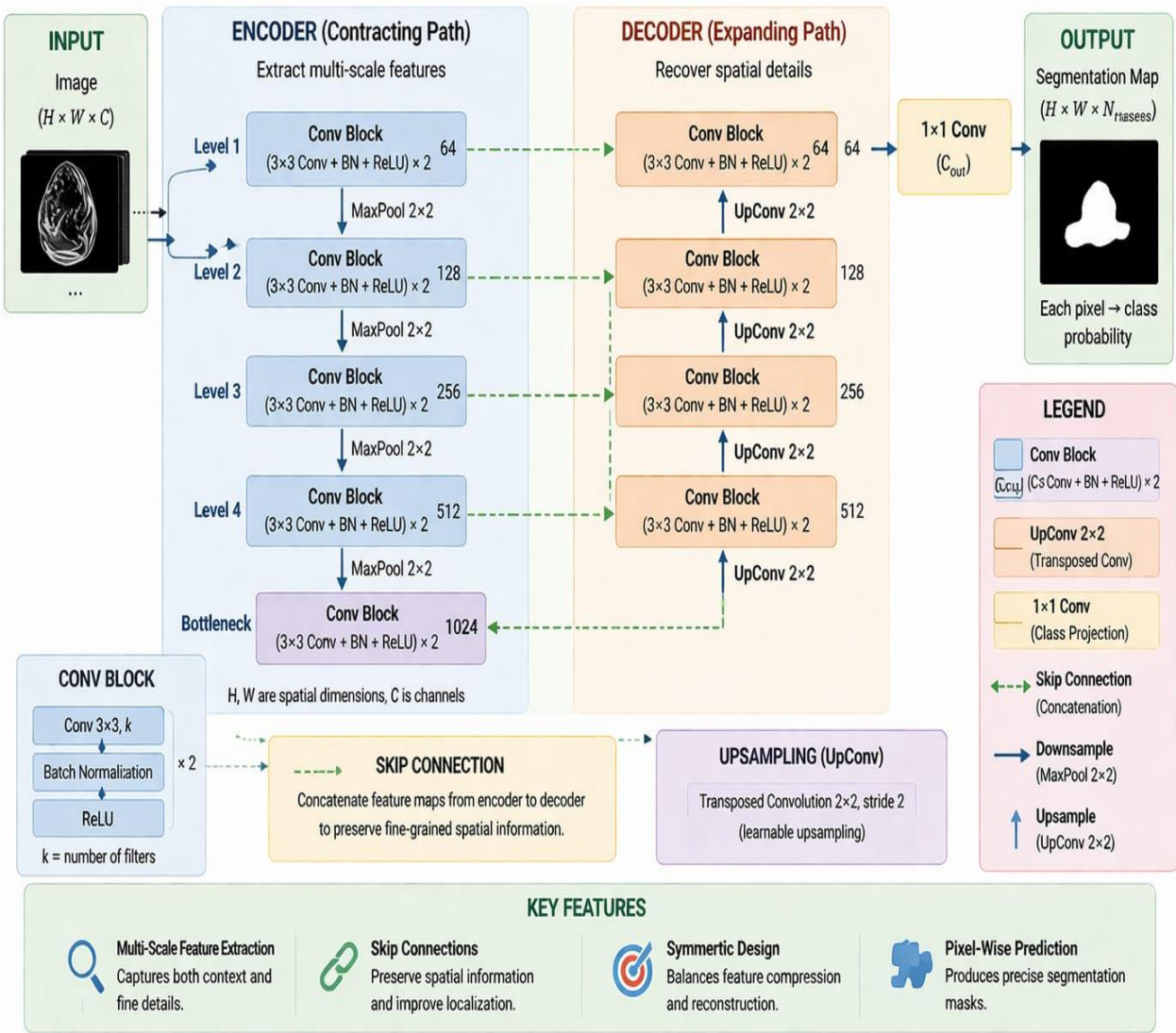
ResNet Block Diagram

The Building Block of Residual Networks



SEGMENTATION NETWORK DIAGRAM

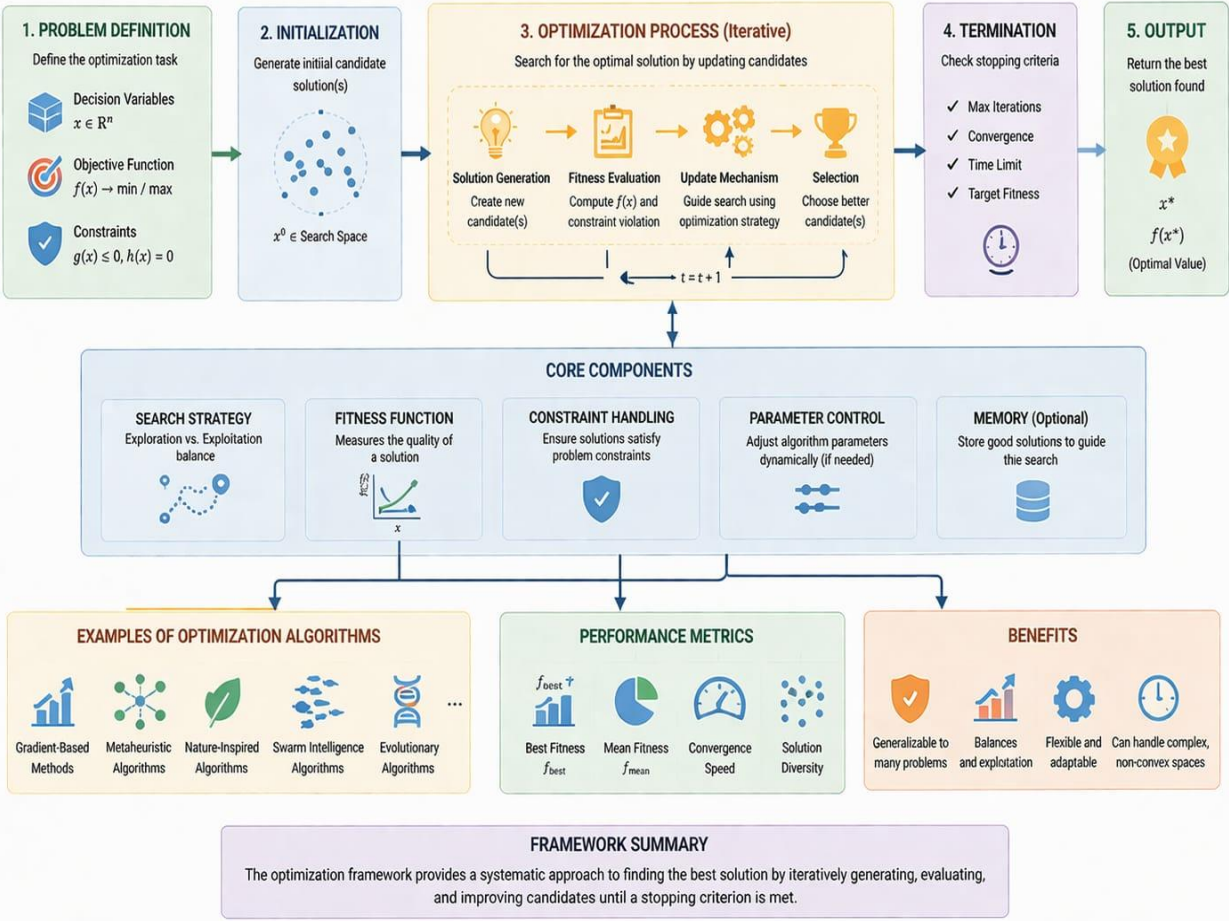
Encoder–Decoder architecture with skip connections for precise pixel-wise segmentation



IDEA: The encoder learns “what” is in the image, and the decoder learns “where” it is, with skip connections bridging the gap to achieve accurate segmentation.

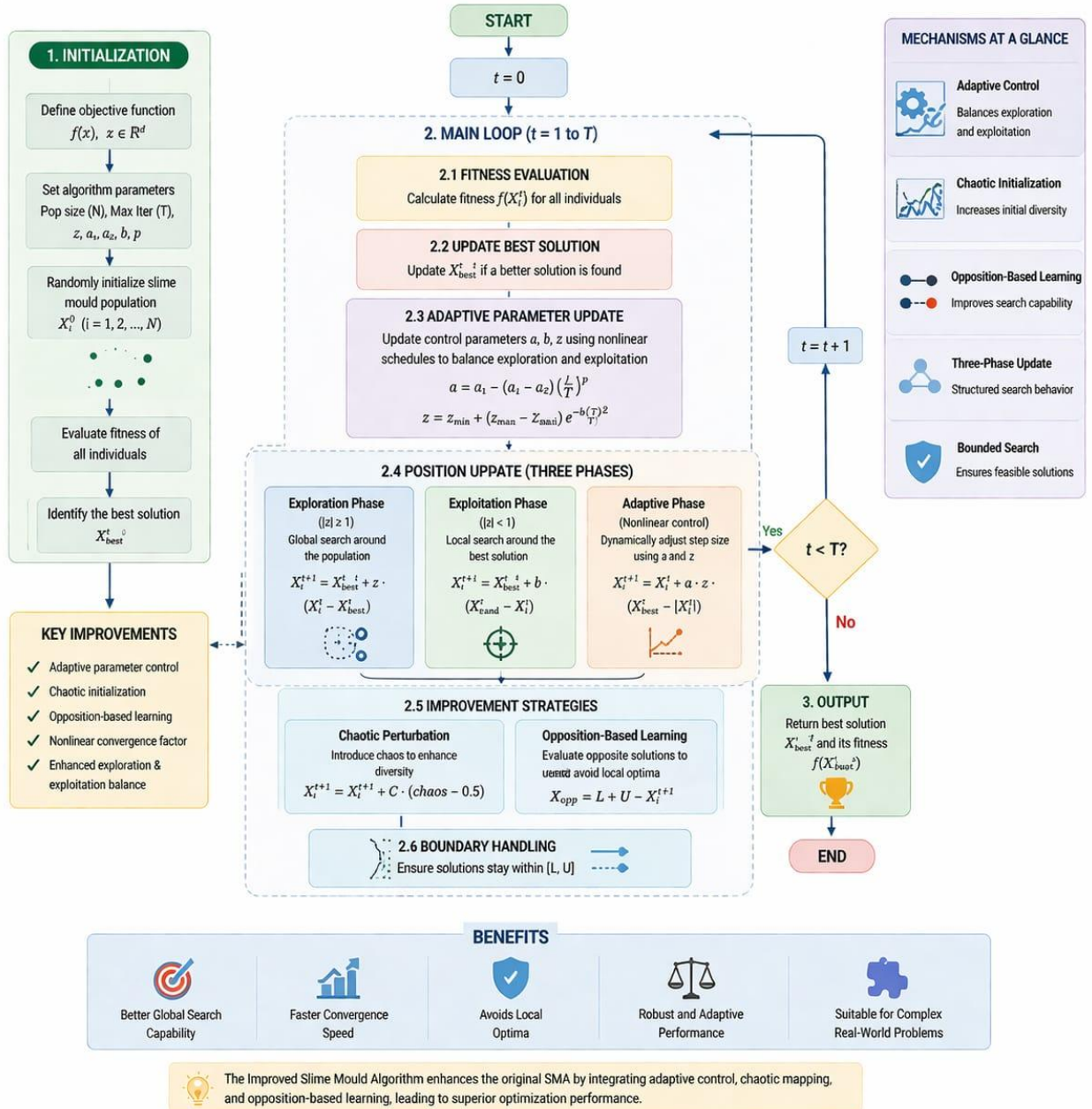
OPTIMIZATION FRAMEWORK DIAGRAM

A Generic Framework for Solving Complex Optimization Problems



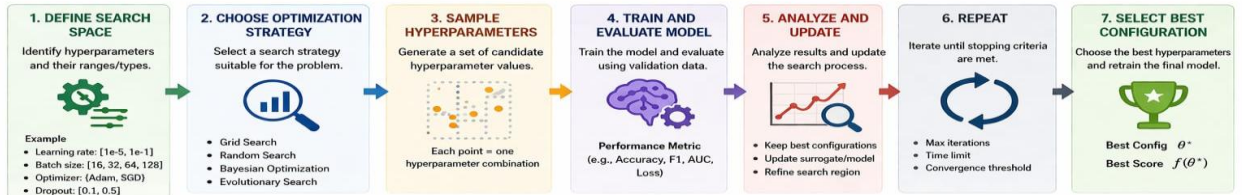
IMPROVED SLIME MOULD ALGORITHM FLOWCHART

An Enhanced Nature-Inspired Optimization Algorithm for Global Search and Fast Convergence



HYPERPARAMETER OPTIMIZATION PROCESS

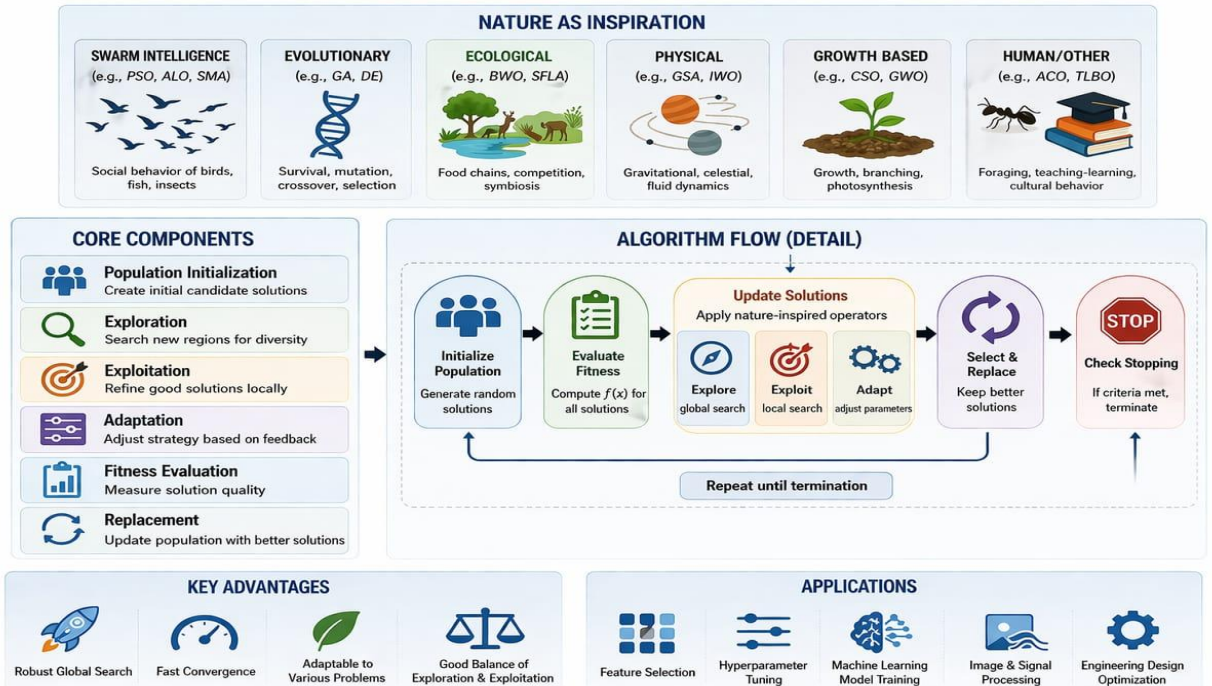
A systematic process to find the best hyperparameters that maximize model performance



Hyperparameter optimization is essential for maximizing model potential and achieving robust generalization.

NATURE-INSPIRED OPTIMIZATION WORKFLOW

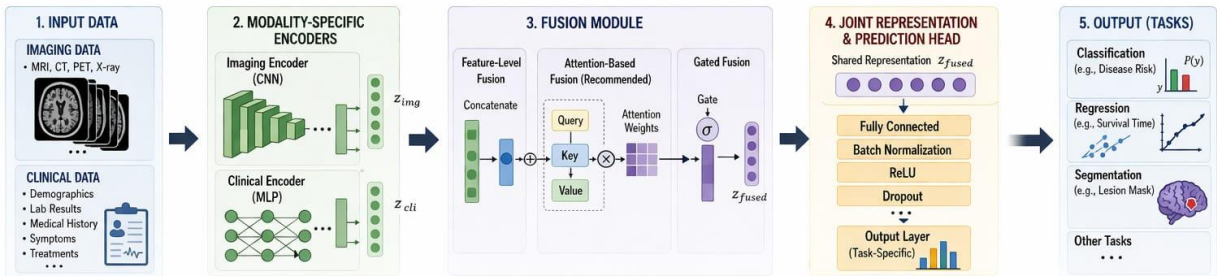
Leveraging Natural Mechanisms to Solve Complex Optimization Problems



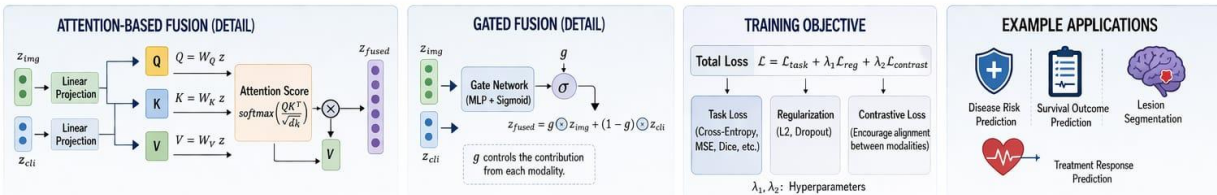
Nature-inspired algorithms mimic natural phenomena to efficiently explore complex search spaces and find high-quality solutions with fewer assumptions about the problem.

CLINICAL + IMAGING DATA FUSION MODEL

Integrating Heterogeneous Data for Improved Predictive Performance



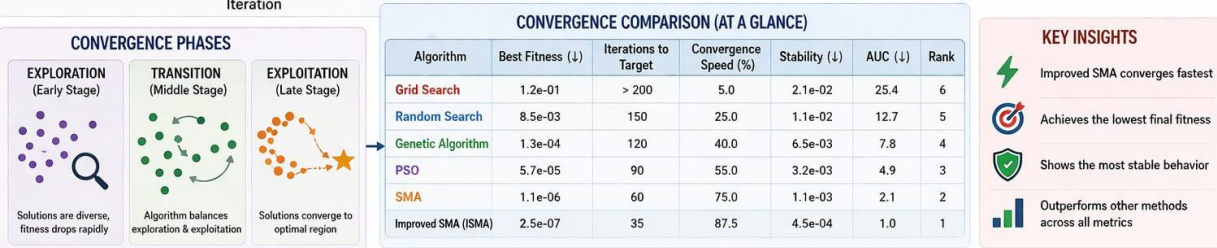
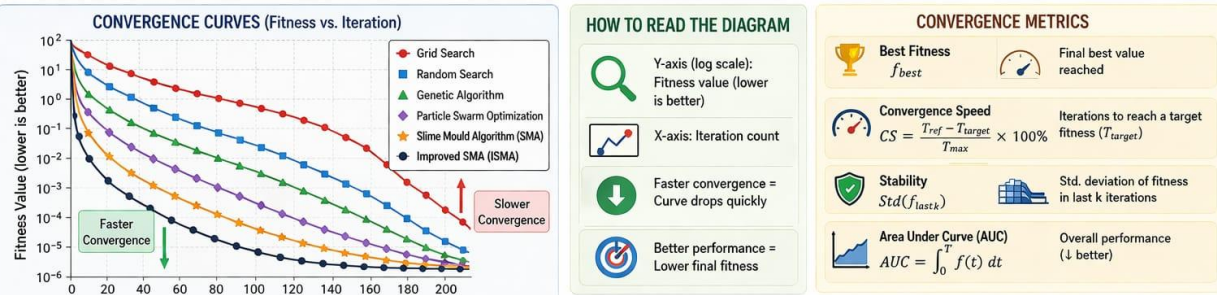
The model learns complementary patterns from imaging and clinical data to make more accurate and robust predictions.



Fusing clinical and imaging data enables the model to capture both visual patterns and patient context, leading to better generalization and real-world impact.

CONVERGENCE BEHAVIOR DIAGRAM

Visualizing and Analyzing the Convergence Patterns of Optimization Algorithms



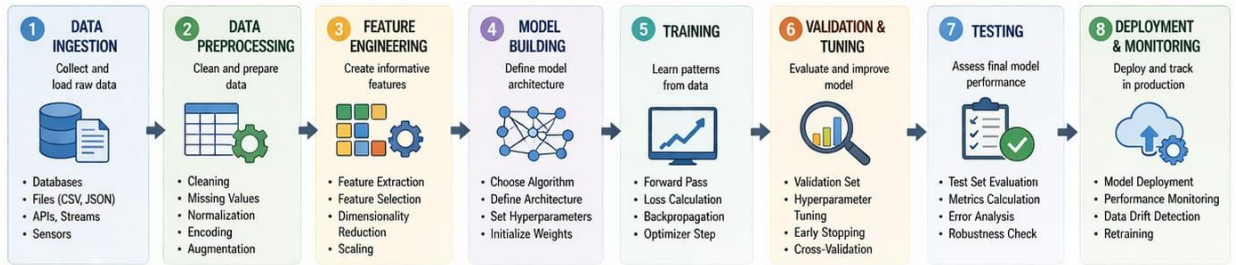
WHY CONVERGENCE BEHAVIOR MATTERS?



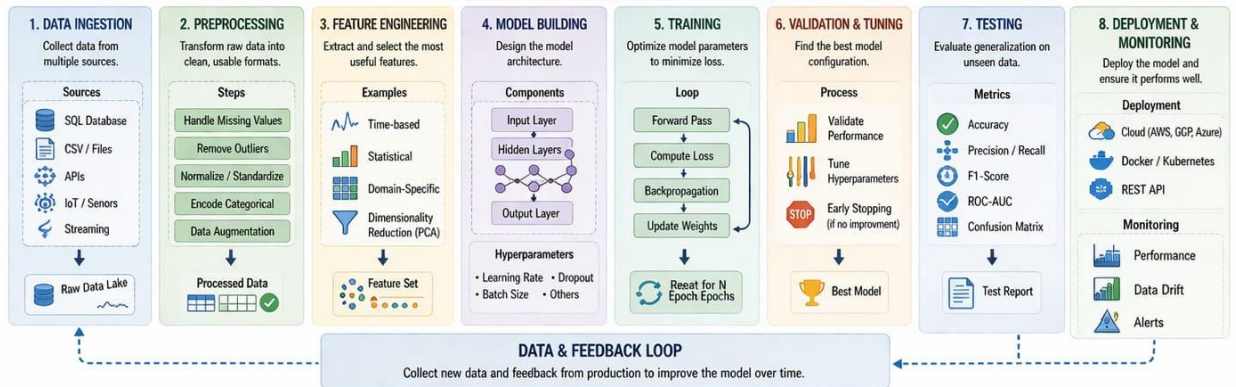
A good algorithm converges fast, reaches a low fitness value, and remains stable.

TRAINING PIPELINE DIAGRAM

An End-to-End Workflow from Data to Deployable Model



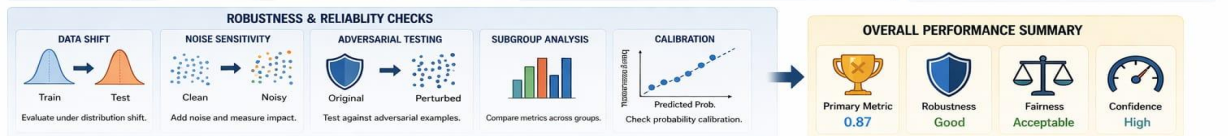
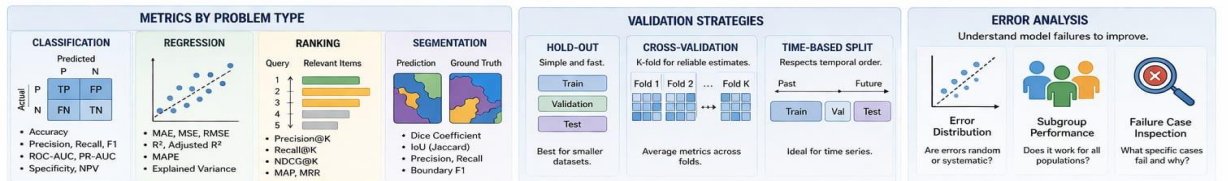
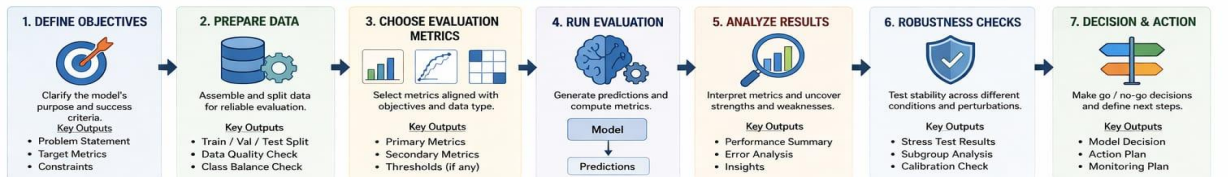
DETAILED VIEW OF EACH STAGE



A well-designed training pipeline ensures data quality, model performance, and reliable deployment—unlocking real-world AI impact.

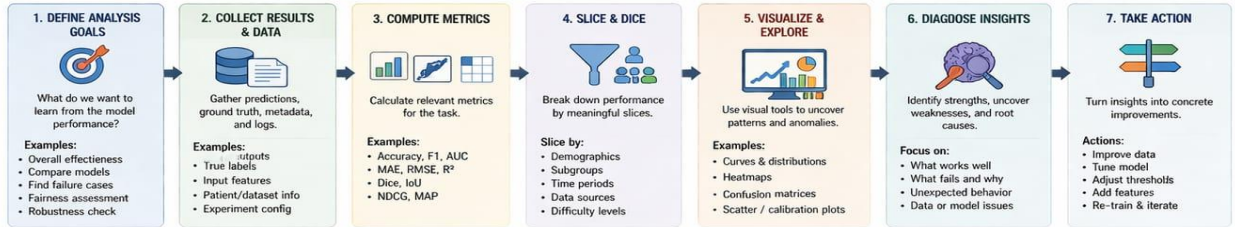
MODEL EVALUATION FRAMEWORK

A Comprehensive Framework for Assessing Model Performance, Robustness, and Reliability

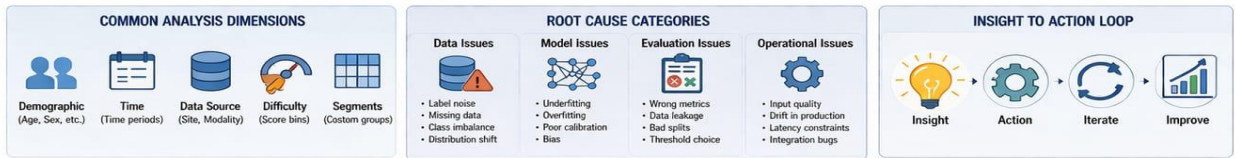
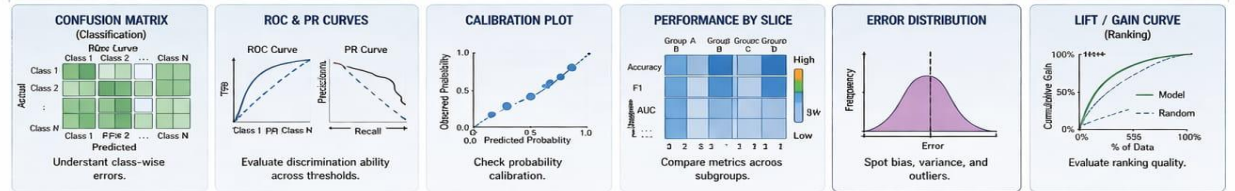


PERFORMANCE ANALYSIS WORKFLOW

A Systematic Approach to Evaluate, Diagnose, and Improve Model Performance



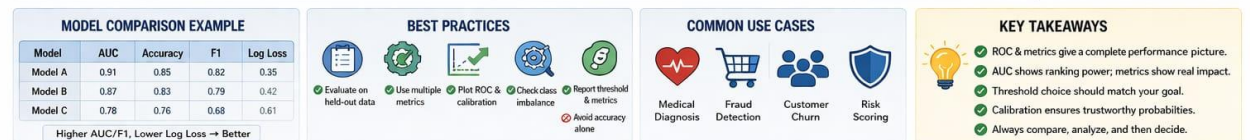
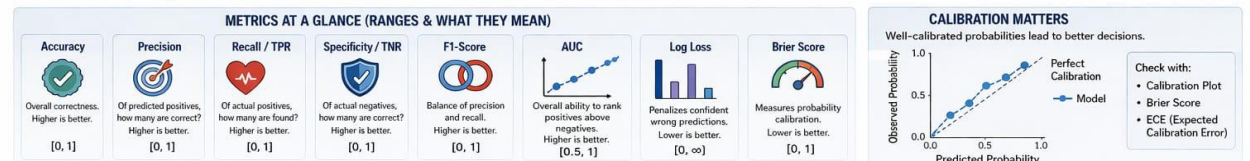
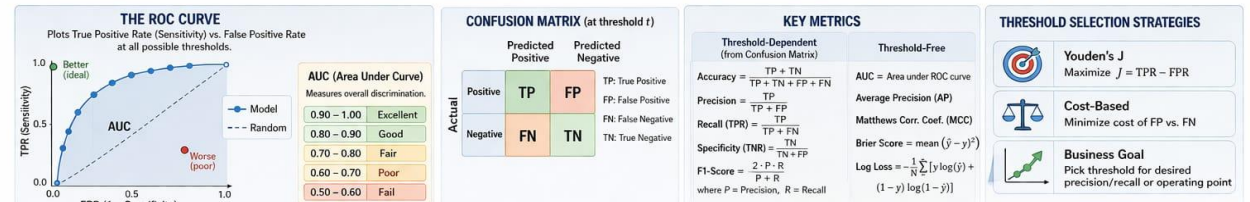
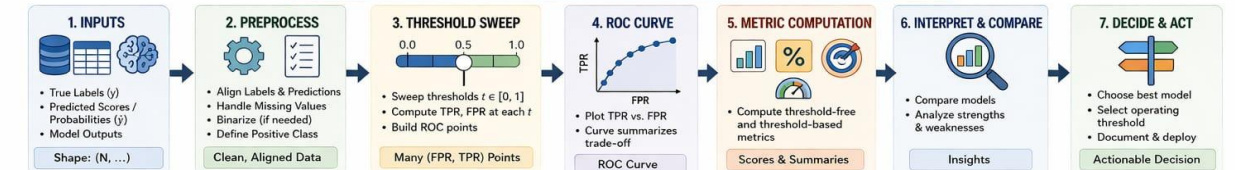
DEEP-DIVE ANALYSIS COMPONENTS



★ Good performance analysis turns numbers into insights—and insights into better models.

ROC & METRICS EVALUATION FLOW

From Predictions to Performance: A Comprehensive Evaluation Pipeline



Form Submitted by:
Name- S Naresh Kumar
Contact No: 9866000203
